

PAPER_158 Hybrid MUGE Blending Model: $g_{\text{hybrid}} = \hat{I}^2 \hat{A} \cdot g_{\text{compressed}} + (1 - \hat{I}^2) \hat{A} \cdot g_{\text{resonance}}$

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Abstract

This paper introduces the **Hybrid MUGE Blending Model**, a continuous interpolation between the Compressed MUGE and Resonance MUGE gravity frameworks. The blending parameter $\hat{I}^2 = \exp(-B/B_{\text{crit}})$ transitions smoothly from $\hat{I}^2 \approx 0$ (pure resonance, magnetar regime) to $\hat{I}^2 \approx 1$ (pure compressed Newtonian+, weak-field regime). This unified hybrid is the **first algebraic bridge** between the Newtonian limit and the full resonance superconductive regime within the UQFF framework, extending Â§2.2 PAPER_155 which proved Newtonian emergence only in the $f_{\text{TRZ}} \rightarrow 0$ limit.

1. Motivation

PAPER_146 (Â§2.2) established the 12-term resonance MUGE master equation. PAPER_090 (Â§1.12) established the 9-term compressed MUGE. Previously, no **analytical bridge** existed for intermediate regimes where B is a significant fraction of B_{crit} (e.g., solar magnetosphere, white dwarfs, neutron star crusts).

The blending model provides: - Continuous function of B/B_{crit} across the full magnetic field parameter space - Smooth limiting behavior at both extremes - Single equation for all astrophysical environments

2. Hybrid Blending Equation

$$g_{\text{hybrid}}(r, t) = \beta \cdot g_{\text{comp}}(r, t) + (1 - \beta) \cdot g_{\text{res}}(r, t)$$

where

$$\beta = e^{-B/B_{\text{crit}}}$$

- g_{comp} = compressed MUGE gravity (PAPER_090, 9-term Newtonian+corrections)
 - g_{res} = resonance MUGE gravity (PAPER_146/159, 12/13-term superconductive resonance)
 - B = local magnetic field strength [T]
 - B_{crit} = critical quantum field (4.4×10^{13} T for magnetars)
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3. Limiting Cases

3.1 Magnetar Regime ($B/B_{\text{crit}} \approx 1$)

$$\beta = e^{-1} \approx 0.368 \Rightarrow g_{\text{hybrid}} \approx 0.37 g_{\text{comp}} + 0.63 g_{\text{res}}$$

For SGR 1745-2900 ($B = 3 \times 10^{13}$ T, $B_{\text{crit}} \approx 4.4 \times 10^{13}$ T):

$$\beta_{SGR} = e^{-3 \times 10^{11} / 4.4 \times 10^{13}} \approx 0.9933 \approx 1$$

g_hybrid g_comp for SGR 1745 (compressed dominant near-critical but below)

3.2 Solar Regime ($B \ll B_{crit}$)

For Sun ($B_s = 10^{-4}$ T):

$$\beta_{Sun} = e^{-10^{-4} / 4.4 \times 10^{13}} \approx 1.000000$$

g_hybrid g_comp (pure Newtonian+ regime)

3.3 Neutron Star Surface ($B \sim 10^{12}$ T)

$$\beta_{NS} = e^{-10^{12} / 4.4 \times 10^{13}} \approx 0.977$$

g_hybrid 0.977 g_comp + 0.023 g_res (dominantly compressed; 2.3% resonance correction)

4. Python Implementation (CP3)

```
import math

def compute_hybrid_muge(g_compressed: float, g_resonance: float,
                        B: float, B_crit: float = 4.4e13) -> float:
    """
    Hybrid MUGE blending: g_hybrid = beta * g_comp + (1-beta) * g_res
    where beta = exp(-B/B_crit).

    Args:
        g_compressed: Compressed MUGE gravity [m/s^2]
        g_resonance: Resonance MUGE gravity [m/s^2]
        B: Local magnetic field [T]
        B_crit: Critical field (default 4.4e13 T, magnetar)

    Returns:
        g_hybrid: Blended MUGE gravity [m/s^2]
    """
    if B_crit <= 0:
        raise ValueError("B_crit must be positive")
    beta = math.exp(-B / B_crit)
    return beta * g_compressed + (1.0 - beta) * g_resonance
```

5. Validation 7-System Blending Table

System	B [T]	$\hat{I}^2$	g_comp [m/s ²]	g_res [m/s ²]	g_hybrid [m/s ²]
SGR 1745-2900	3×10^{11}	0.9933	1.783×10^3	1.655×10^{-4}	1.785×10^3 + dom

